Saving money is not the only motivation...

For many, open source technology is all about saving on hefty IT budgets. Well, Goibibo thinks a bit differently. The use of open source technology at the travel aggregator has more to do with technology than just financial benefits. Sahni shares, "We are running a very effective Big Data cluster over open source Hadoop components, which not only saved money but gave us a platform to embrace other open source tools like Hive, Hbase, etc, to scale our data mining process. A data source like Redis helped us in churning and storing large data structures for our real-time application needs, resulting in improving Goibibo's performance multi-fold.

"Apart from the technological benefits, one cannot deny the financial benefits that open source technology offers. I can't say what the exact numbers are when it comes to slashing the IT budget, but I would say it is a fairly substantial saving that we have achieved through the usage of open source technology."

There are challenges too!

Choosing an unusual path definitely comes with its share of challenges. Goibibo's open source journey came with its share of hardships too. But Sahni's team knows how to deal with them. He shares, "Only when we use open source technology that is not too widely used nor popular, but solves our specific use cases, do we face challenges in terms of documentation and support, though this is not the case with most popular open source projects.

"To overcome these challenges we work very closely with the developers and maintainers of that code, through IRC channels, mails or forums to ensure we get going in the right direction. And while doing this, we also help in testing that particular technology."

Yet another challenge faced by Sahni is getting the right talent. He says, "Although open source is getting popular and we have started seeing emerging talent on popular technology stacks, yet, it is still a challenge getting the right candidates for technologies that are fairly new." To counter the problem, the company focuses on hiring engineers with the right aptitude, and then lets them dabble with the technologies and come up with solutions to their specific technology needs.

The beauty of open source is that it is backed by a strong community and the tech team at Goibibo knows how to make the most of it. Team members work very closely with the community, and this is one of the main reasons behind their love for open source. Sahni says, "Whenever we face challenges, we try to talk with the relevant folks over forums and IRC channels, and also over email groups, at times. During one difficult situation, the committer and owner of the *mod_uwsgi* project (a module over Apache that we use to serve and execute our Python Web application) helped us with the best practices when using the module on a large scale. Also, the responses to queries are pretty fast and in a good number of cases, the turnaround time has been on par or quicker than what proprietary software guarantees."

Making the most of community contributions is what Goibibo is best at, but it doesn't end there. The tech team is passionate about the art of sharing too. Sahni says, "We have our organisation's Github page and we regularly push code that we want to open source. And our engineering team works closely on some open source projects. Also, some engineers in Goibibo own certain fairly popular open source projects as well."

Way to go, Goibibo! END

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Source file

The source file can be hosted either on the Salt master server or on a network accessible server. The file hosted on a network server (e.g., HTTP or FTP) requires the *source_hash* argument. We could also keep the data files along with *init.sls* but it is easier to maintain and manage these files if they are separated from the SLS configuration. If we do keep the data and configuration files within the same directory, the source statement and files could look like what's shown below:

```
nginx/init.sls
nginx/nginx.conf

- source: salt://nginx/nginx.conf
```

Debugging

Once the rules in an SLS are ready, they should be tested to ensure they work properly. To invoke these rules, execute *salt '*' state.highstate* on the command line. If you get back only hostnames followed by a ':', but no return, chances are that there is a problem with one or more of the SLS files. On the minion, use the command *salt-call state.highstate -l debug* to examine the output for errors. This should help to troubleshoot the issue. The minions can also be started in the foreground in debug mode: *salt-minion -l debug*. END

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